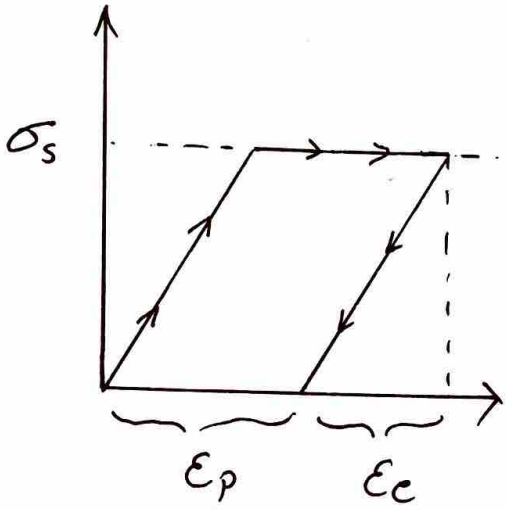
Övning 5 - Termoelasticitet, plasticitet och elastisk vridning

2.2: 22, 38 2,6: 12, 14

Plasticering



ε_p - Plastisk töjning

ε_e - Elastisk töjning

P_s - Last vid initiell plasticering

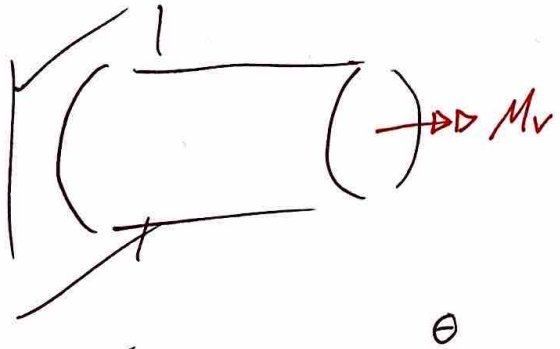
P_f - Kollapslast, last vid full plasticering

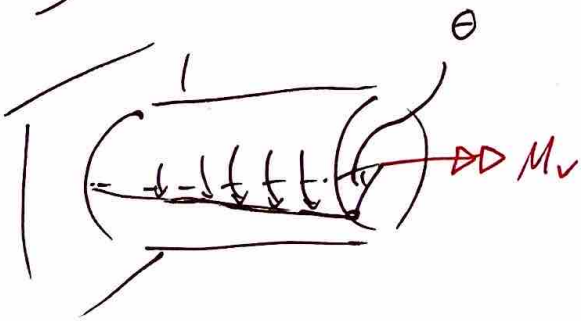
Flytlast förhöjning

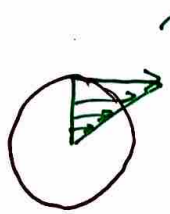
$$\beta = \frac{P_f - P_s}{P_s}$$

Procentuell ökning från initiell plasticering till kollapslast (P_f)

Vridning







[FS. 6.76]

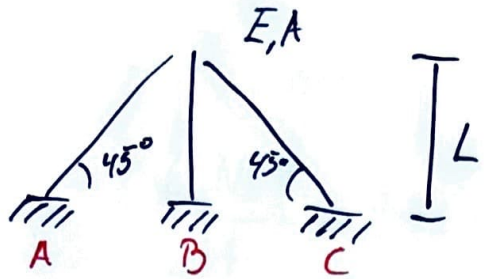
$$\tau_{max} = \frac{M_v}{W_v}$$

W_v - Vridmotståndet, olika för olika tvärsnitt ($W_v = \frac{\pi}{2b}(b^4 - a^4)$ För rör)

θ - Förvridningen

$$\theta = \frac{M_v L}{G I_k}$$ [FS. 6.75]

2.2.22



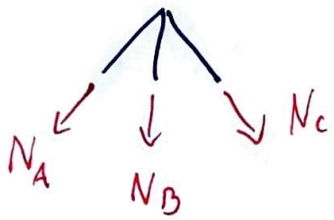
Temperaturen höjs ΔT

Givet: Radie a
E-modul E
Utvidgningsfaktor α

Sökt: Normalspänning i mittstång σ_B

Jämvikt

① Frilägging



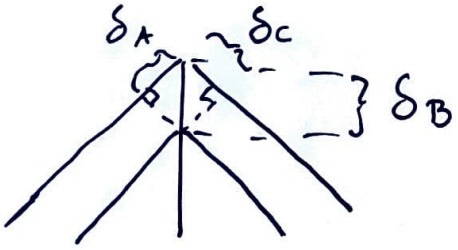
$\rightarrow$: $N_A = N_C$

$\uparrow$: $-N_B - N_A \sin\alpha - N_C \sin\alpha = 0$

$$N_B + 2N_A \frac{\sqrt{2}}{2} = 0$$

$$\boxed{N_B + \sqrt{2} N_A = 0}$$

② Deformationssamband



$\delta_A = \delta_C$

$$\underbrace{\sin\alpha}_{\frac{\sqrt{2}}{2}} = \frac{\delta_A}{\delta_B} \Rightarrow \boxed{2\delta_A = \sqrt{2}\delta_B}$$

③ Deformation

$$\delta = \frac{NL}{EA} + \alpha \Delta T L \Rightarrow \begin{cases} \delta_A = \dfrac{\sqrt{2} N_A L}{EA} + \alpha \Delta T \sqrt{2} L \\ \delta_B = \dfrac{N_B L}{EA} + \alpha \Delta T L \end{cases}$$

2.2.22 del 2

Insatt i ekv (II)

$$\frac{2\sqrt{2}N_A L}{EA} + 2\sqrt{2}\alpha\Delta T L = \frac{\sqrt{2}N_B L}{EA} + \sqrt{2}\alpha\Delta T L$$

$$\Rightarrow N_B = 2N_A + \alpha\Delta T EA$$

Insatt i ekv (I)

$$2N_A + \alpha\Delta T EA + \sqrt{2}N_A = 0$$

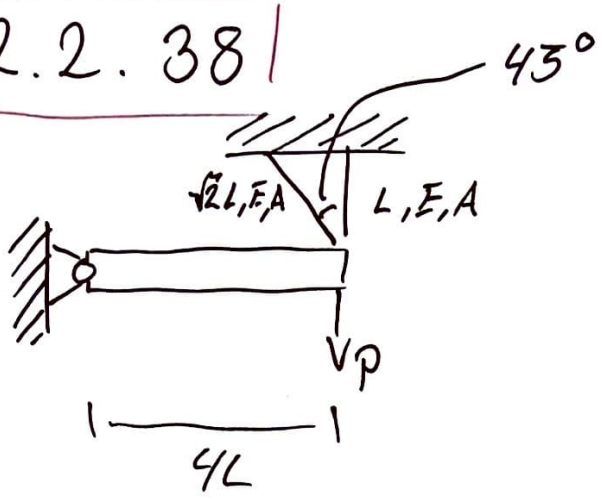
$$N_A = -\frac{\alpha\Delta T EA}{2+\sqrt{2}}$$

$$N_B = \frac{\alpha\Delta T EA}{\sqrt{2}+1}$$

④ spänning

$$\boxed{\sigma_B = \frac{N_B}{A} = \frac{\alpha\Delta T E}{\sqrt{2}+1}}$$

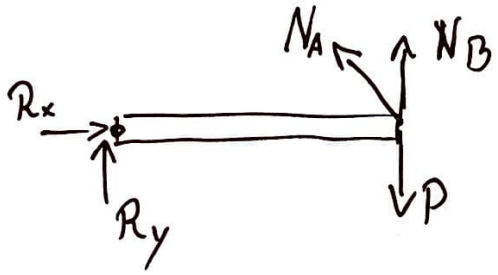
2.2.38



Givet: L, E, A, σ_s

Sökt: Spänningarna
σ_A, σ_B
Vertikala förskjutning
δ
Flytlast förhöjning
β

① Frilägging



Jämvikt

$\rightarrow: \quad R_x - N_A \frac{\sqrt{2}}{2} = 0$

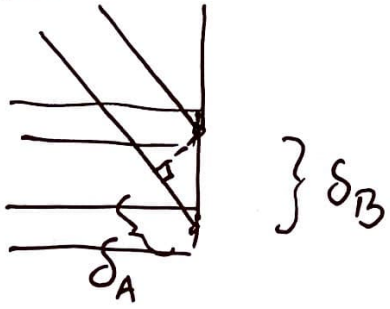
$\uparrow: \quad R_y + N_B + N_A \frac{\sqrt{2}}{2} - P = 0$

$\circlearrowright \quad 4PL - 4N_B L - 4\frac{\sqrt{2}}{2} N_A L = 0$

$$\Rightarrow \quad N_B + \frac{\sqrt{2}}{2} N_A = P$$

1 ekv
2 okända

② Deformation



$$\sin\alpha = \frac{\delta_A}{\delta_B} \quad \Rightarrow \quad \delta_B = \sqrt{2}\,\delta_A$$

③ Deformation

$$\delta_A = \frac{\sqrt{2} N_A L}{EA} \qquad \delta_B = \frac{N_B L}{EA}$$

$$N_B = 2N_A \quad \Rightarrow \quad N_A = \frac{2P}{4+\sqrt{2}} \qquad N_B = \frac{4P}{4+\sqrt{2}}$$

2.2.38 del 2

$$\sigma_A = \frac{2}{4+\sqrt{2}}\frac{P}{A} \qquad \sigma_B = \frac{4}{4+\sqrt{2}}\frac{P}{A}$$

Stång 2 Plasticerar först

$$\delta_B = \frac{4}{4+\sqrt{2}}\frac{PL}{EA}$$

④ Flytlast förhöjningen

P_s bestäms från första plasticering vilken sker i stång B

$\sigma_B = \sigma_s$ ← Viktigt!

$$\Rightarrow \sigma_s = \frac{4}{4+\sqrt{2}}\frac{P_s}{A} \qquad P_s = \frac{4+\sqrt{2}}{4}\sigma_s A$$

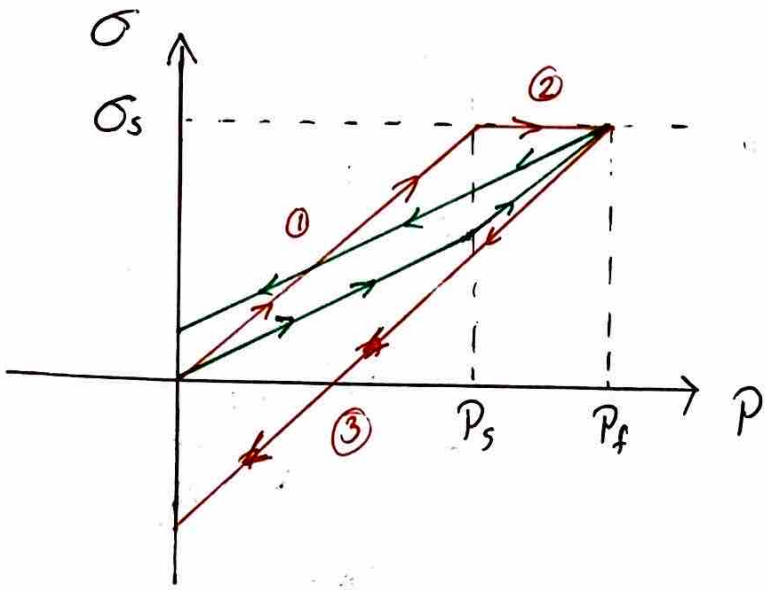
P_f bestäms när alla stänger uppnår plasticering $\sigma_A = \sigma_B = \sigma_s$, $N_A = N_B = \sigma_s A$

$$\sigma_s A + \frac{\sqrt{2}}{2}\sigma_s A = P_f \qquad P_f = \frac{2+\sqrt{2}}{2}\sigma_s A$$

$$\beta = \frac{P_f - P_s}{P_s} = 0.27$$

2.2.38 del 3

b) Vad är spänningarna σ_A, σ_B efter plastisk kollaps



① På lastning till initiell plasticering (P_s)

② Pålastning till kollapslast (P_f)

③ Avlastning elastiskt

Stång B Stång A

Vid full kollaps gäller att alla stänger uppnått sträckgränsen

$$\sigma_A^f = \sigma_s \qquad \sigma_B^f = \sigma_s$$

Avlastning sker enligt elastiska samband från kollapslast

$$\sigma_A^a = \frac{2}{4+\sqrt{2}}\frac{P}{A} \qquad \sigma_B^a = \frac{4}{4+\sqrt{2}}\frac{P}{A}$$

(svar som står längst upp sida 2)

Avlastning sker enligt addition

$$\sigma_A^R = \sigma_A^f + \sigma_A^a(-P_f) \qquad \sigma_B^R = \sigma_B^f + \sigma_B^a(-P_f)$$

$$\sigma_A^R = \sigma_s - \frac{2}{4+\sqrt{2}}\frac{P_f}{A} = \sigma_s - \frac{2}{4+\sqrt{2}}\frac{1}{A}\frac{\sqrt{2}+2}{2}\sigma_s A = \frac{2}{4+\sqrt{2}}\sigma_s$$

$$\sigma_B^R = \sigma_s - \frac{4}{4+\sqrt{2}}\frac{P_f}{A} = \sigma_s - \frac{4}{4+\sqrt{2}}\frac{1}{A}\frac{\sqrt{2}+2}{2}\sigma_s A = -\frac{\sqrt{2}}{4+\sqrt{2}}\sigma_s$$

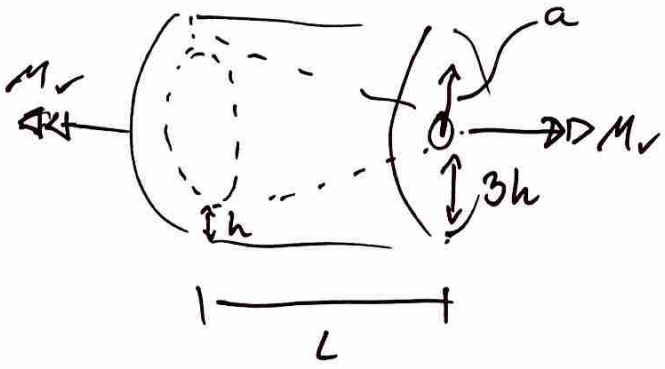
$$\boxed{\sigma_A^R = \frac{2}{4+\sqrt{2}}\sigma_s} \qquad \boxed{\sigma_B^R = -\frac{\sqrt{2}}{4+\sqrt{2}}\sigma_s}$$

2.6.12

Skjuvmodul G

Sökt: Förvridningen Θ

(Atag tunnväggigt, $a >> h$)

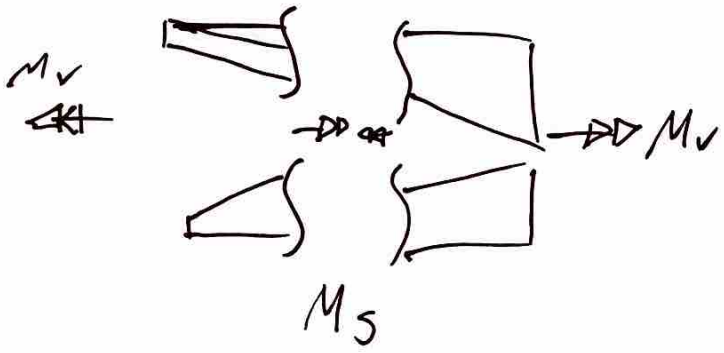


① skiftning

Jämvikt

$\rightarrow\!\!\!\!\rightarrow$: $M_v - M_s = 0$

Samma snittmoment genom hela stången.



② Förvridning

$$\Theta = \int_0^L \vartheta \, dx$$

Θ - totala förvridningen

ϑ - vridning per l.e.

$$\vartheta = \frac{M_v}{G K(x)} \qquad K(x) = 2\pi a^3 t(x) \qquad [FS: 6.80]$$

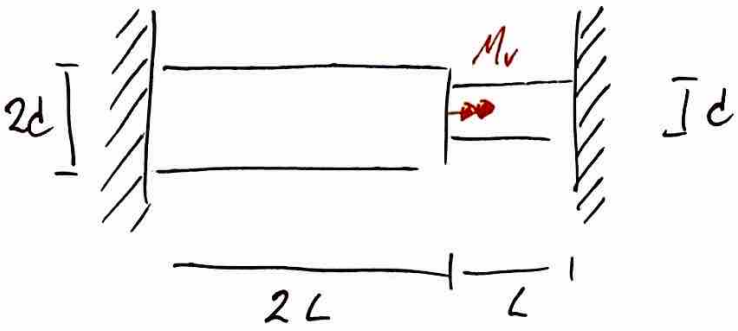
$$t(x) = h + \frac{2h}{L} x$$

$$\vartheta = \frac{M_v}{G 2\pi a^3 \left(h + \frac{2h}{L} x\right)}$$

$$\Theta = \int_0^L \frac{M_v}{2G\pi a^3 h \left(1 + \frac{2}{L} x\right)} dx = \frac{M_v}{2G\pi a^3 h} \int_0^L \frac{1}{1 + \frac{2}{L} x} dx$$

$$= \frac{M_v L}{4 G \pi a^3 h} \ln(3)$$

2.6 : 14

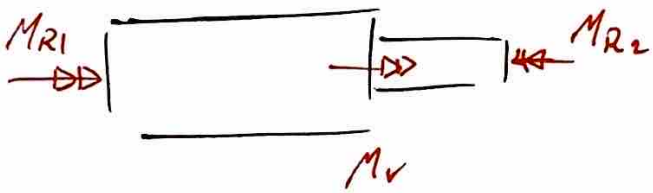


Givet: d, L, τ_{till}

Sökt: Maximalt vridmoment då någon del av strukturen uppnår τ_{till}

$M_{v,max}$

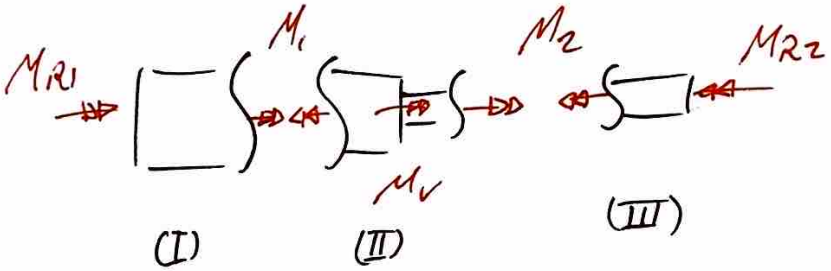
① Frilägganing



Jämvikt

$$\rightarrow\!\!\!\rightarrow \; M_{R1} + M_v - M_{R2} = 0$$

② Skiftning



(II) $\rightarrow\!\!\!\rightarrow$ $M_v + M_2 - M_1 = 0$ (1)

1 ekvation
2 okända

③ Deformationssamband

$$\theta_{tot} = \theta_1 + \theta_2 = 0 \quad (2)$$

$$\theta = \frac{M_v L}{GK} \; [FS\,6.75] \qquad K = \frac{\pi D^4}{32} \; [FS\,6.77]$$

$$\Rightarrow \quad K_1 = \frac{\pi d^4}{2} \qquad \theta_1 = \frac{4 M_1 L}{G \pi d^4}$$

$$K_2 = \frac{\pi d^4}{32} \quad \Rightarrow \quad \theta_2 = \frac{32 M_2 L}{G \pi d^4}$$

θ_1 och θ_2 insatt i ekv (2)

$$\frac{4L}{G\pi d^4}(8M_2 + M_1) = 0 \quad \Rightarrow \quad M_1 = -8M_2$$

M_1 insatt i ekvation (1)

$$M_v + M_2 - (-8M_2) = 0$$

$$\Rightarrow \quad M_2 = -\frac{1}{9}M_v \qquad M_1 = \frac{8}{9}M_v$$

④ Plasticering

$$\tau_{max} = \frac{M_v}{W_v} \text{ [FS 6.76]} \qquad W_v = \frac{\pi D^3}{16} \text{ [FS 6.78]}$$

$$\tau_{max1} = \frac{8}{9}M_v \frac{16}{\pi(2d)^3} \Rightarrow \tau_{max1} = \frac{16}{9}\frac{M_v}{\pi d^3}$$

$$\tau_{max2} = -\frac{1}{9}M_v \frac{16}{\pi d^3} \Rightarrow \tau_{max2} = -\frac{16}{9}\frac{M_v}{\pi d^3}$$

Båda stängerna plasticerar samtidigt

$$|\tau_{max1}| = |\tau_{max2}| = \tau_{till}$$

$$\tau_{till} = \frac{16}{9}\frac{M_{v,max}}{\pi d^3} \Rightarrow \boxed{M_{v,max} = \frac{9}{16}\pi d^3 \tau_{till}}$$